JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

SEPTEMBER 2021 EXAMINATIONS

PHYSICS PRACTICAL **Time Allowed: 2½ Hours**

Answer All Questions.

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Question 1

APPARATUS: Perforated metre rule, Split cork, Retort stand and clamp, One optical pin, Stop watch

PROCEDURE

(a) You are provided with a T-shaped card/metre rule. The card/metre rule has a mark on it and 14 holes. Label the holes on the card/metre rule, 1 to 14, starting from the top end as shown in Fig. 1.1.

(b) (i) Set up the apparatus, placing the pin through hole number 9, as shown in Fig. 1.1.

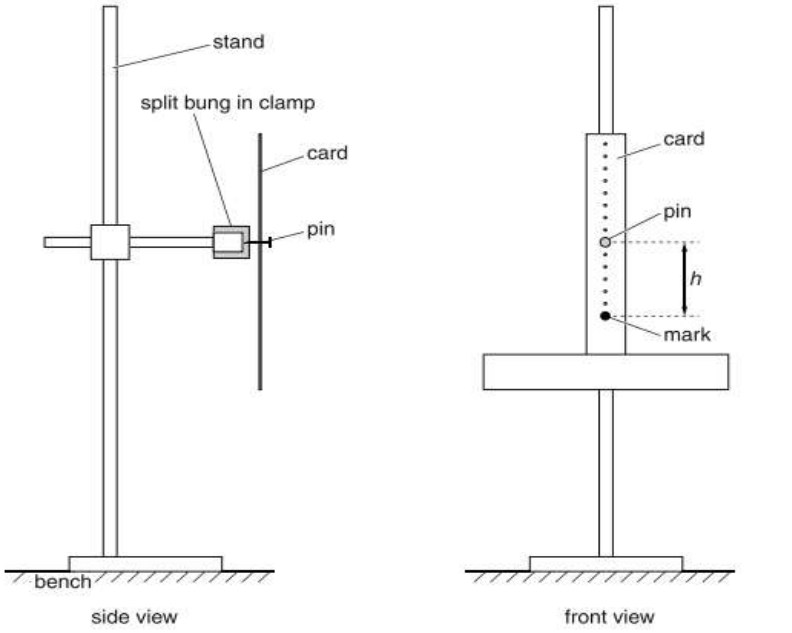


Fig. 1.1

(ii) Measure and record the distance *h* between the pin and the mark.

(c) Displace the card shape/metre rule to the left. Release the shape and watch its movement. The shape will move to the right and then to the left again, completing a swing as shown in Fig. 1.2.

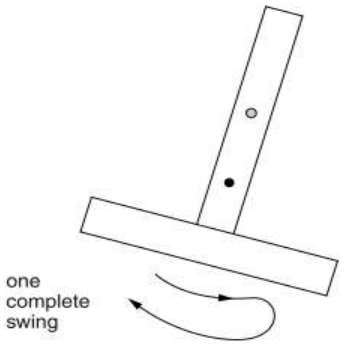


Fig. 1.2

Measure and record the time for at least 10 swings.

Record enough readings to determine an accurate value for the time *T* taken for one complete swing.

(d) Vary *h*, by changing the hole through which the pin is placed upward, and repeat (b)(ii) and (c) until you have six sets of values of *h* and *T*.

Include values of T^2h and h^2 in your table.

(e) (i) Plot a graph of T^2h on the y-axis against h^2 on the x-axis.

(ii) Draw the straight line of best fit.

(iii) Determine the gradient and y-intercept of this line.

The quantities T and h are related by the equation

$$T^2h = Ph^2 + Q$$

Where P and Q are constants.

Using your answers from (e) (iii), determine the values of P and Q.

Give appropriate units. **[TOTAL = 10 marks]**

Question 2

APPARATUS: Two cells of 1.5 V each, high resistance voltmeter (0 – 5V), ammeter (0 – 1A), variable resistor, connecting wires and a key.

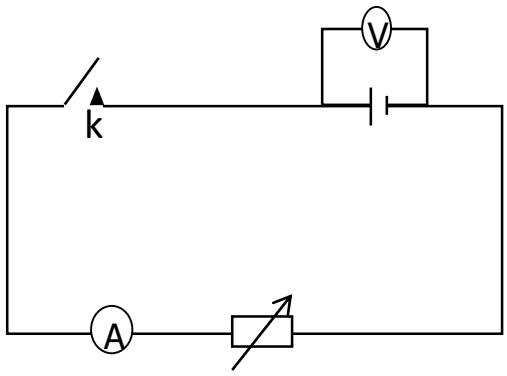


Fig. 2

PROCEDURE

(i) Set up the circuit as shown in Fig. 2.

(ii) With the key opened, record the reading of the voltmeter connected across the cell. Record this Value as E.

(iii) Close the key and adjust the variable resistance so that a small current passes through the circuit. Record this current (I) and the corresponding voltmeter reading (V).

(iv) Keep the key closed and adjust the variable resistance to obtain a slightly larger current. Record the value of this current I and the corresponding voltmeter reading V.

(v) Repeat the preceding step to obtain 6 sets of readings of I and V.

(vi) Record your results as shown in table below.

V(V)						
I(A)						
E – V(V)						

Questions

(a) Plot a graph of (E – V) against *I*

(b) Determine the slope of the graph.

(c) State the relationship between E, V, I and r where r is the internal resistance of the cell.

(d) Hence determine the internal resistance, r of the cell.

(e) Why do we use a high resistance voltmeter in this experiment?

(f) State two precautions you took to ensure accuracy.

[TOTAL = 10 marks]